Fiscal Year:	FY 2008	**Task Last Updated:**	FY 05/20/2009
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Project Title:	Mechanism of Radiation-Induced Alteration of Astrocyte-Neuronal Coupling		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	PI is deceased as of January 2010.		
Project Type:	Ground	**Solicitation / Funding Source:**	2006 Radiation Biology NNJ06ZSA001N
Start Date:	10/01/2006	**End Date:**	09/30/2009
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Nelson, Gregory (Loma Linda University)		
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			

Neurons and astrocytes form an interconnected network; a syncytium in which an alteration in one cellular component inevitably affects the other. Disturbances in neuron-astrocyte interactions appear to be at the center of several neurological disorders such as cerebral ischemia, hepatic encephalopathy, and neurodegeneration. The current literature now reflects this understanding with research exploring various metabolic and signaling aspects of this coupling. A significant portion of neuroglial coupling is mediated via glutamate uptake and metabolism, and involves the recycling of substrates between neurons and astrocytes (e.g. glutamate-glutamine shuttle and lactate shuttle). Thus, metabolic and functional cooperativity between astrocytes and neurons is dependant on glutamate. Our recent preliminary data suggests that astrocytes and neurons have reciprocal functional responses to irradiation and very different recovery kinetics. This data has led to the following hypothesis.
Objective: To better understand the mechanistic changes occurring in the astrocytes-neuron couple following exposure to

Task Description:	radiation. We hypothesize that ionizing radiation disrupts the metabolic coupling between neurons and astrocytes by altering glutamate transport and metabolism. We propose to investigate the mechanism for this uncoupling by measuring alterations in glutamate uptake and metabolism, and to determine how abnormal glutamate uptake affects intercellular signaling following irradiation. To achieve these objectives we will utilize NTera2/D1 cells that can be induced to differentiate into mature neurons and astrocytes. These differentiated cell types will be used to investigate the mechanism of neurometabolic uncoupling following acute exposure to 290 MeV carbon and 1 GeV iron ions. Note- the gamma and proton exposure sets are underway and will be supported by intramural funding. The objectives will be achieved by the following specific aims. Specific Aim 1: Measure glutamate transport and metabolism in human astrocytes and neuronal cultures following acute exposure to 290 MeV carbon and 1GeV iron ions delivered as single fractions of 0, 10, 50 and 200 cGy. Samples will be harvested at 3 hrs, 2 and 7 days post irradiation. This aim will also explore the physical and soluble dependency of homeostatic cooperativity between astrocytes and neurons. Estimated time to complete Aim 1 is 24 months with 2 trips to the NSRL at BNL. Specific Aim 2: Measure the endogenous and temporal changes in expression of antioxidants and reactive oxygen species in astrocytes and neurons using the same radiation conditions described in aim 1. Estimated time to complete Aim 2 is 24 months and 3 trips to the NSRL (concurrent with aim 1). Specific Aim 3: Examine alterations in inter-astrocytic and neuronal communication after exposure to ionizing radiation by analyzing essential elements of physical coupling and calcium signaling. Estimated time to complete Aim 3 is 12 months with 2 trips to the NSRL at BNL. The effects of radiation on the central nervous system (CNS) are an issue that needs further investigation especially in view of the increased potential to be exposed to radiation, such as, nuclear reactor accidents, terrorism, radiotherapy, and space travel. Of particular interest are the acute and late effects on the CNS following exposure to radiation, and the underlying mechanism(s) controlling these changes? Preliminary findings indicate that exposure to charged particles results in changes in behavior, cognition, and motor skills. These effects are similar to those seen in neurodegenerative disorders. While there are studies that describe generalized cellular and anatomical changes in the CNS after exposure to radiation, the cellular and molecular mechanisms are unknown. The goal of the proposed project addresses the underlying mechanistic basis for radiation-induced alteration in the coupling between astrocytes and neurons. Our goals correspond to one of NASA's critical questions, specifically to "Acute and Late CNS Risks" from the critical path roadmap (risk 29). The Critical Path Roadmap system for radiation effects on the central nervous system is Risk #29 -- Acute and Late CNS Risks. All of the research and technology questions associated with this risk have priority level one for Mars missions and priority level 3 levels for ISS and 30d lunar missions. The questions addressed by our NASA grant are 29a (functional changes), 29b (late degeneration), 29g (cell and tissue mechanisms of damage) and 29h (different cell populations including stem cells). Additionally, the recent NASA workshop roundtable on CNS radiobiology identified supporting questions to assist in designing experiments to elucidate CNS effects in response to radiation exposure. These questions include: 1) What is the latency to CNS injury, and 2) What cell types are involved in CNS injury (neurons, stem cells, glia)? We are using this framework to focus our investigation of astrocyte neuronal coupling in response to charged particles irradiation. Specifically we are quantifying cellular and functional changes in the glutamate uptake and inter astrocytic-neuronal reactions following 290 MeV carbon and Fe-56 particles. We will use our in vitro culture system with sufficient statistical replicates to address workshop questions 1 and 2 by examining the responses of different constituent cell types and the latency of damage. Aim 1-3 quantifies coordinated responses between two different cell populations (astrocytes/neurons) the effect of radiation induced ROS and the requirements of physical coupling (29a, 29b, 29g, 29h). Our approach is therefore aligned with the programmatic goals set forth by NASA and its advisory groups.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The response of the central nervous system to ionizing radiation will have a direct impact on long duration space flight, and to a lesser degree radiotherapy. By investigating the subtleties of soluble and physical coupling between the critical cells of the CNS (neurons and astrocytes) we will be contributing to the understanding of how low fluence HZE particles present in space will impact astronaut wellbeing. Thus helping to define the limits via biological constraint on such exposures. The results of this NASA sponsored project will quantify cell loss and tissue functional changes from carbon and iron irradiation that can be used to estimate normal tissue complication probabilities in the context of extended space exploration.
	This project was funded by NASA through the NASA cooperative agreement with Loma Linda University. It initiated October first 2006. The first manuscript will be the complete characterization of the cultures and the function of the glutamate transporters in response to gamma radiation. These findings were presented at the recent International Congress of Radiation Research in San Francisco (July 8-12th, 2007). In March of 2007 we irradiated our isolated neurons and astrocytes as well as physically mixed neurons and astrocytes (1:10) with carbon ions at the NSRL (run 7A). The initial findings discussed later in this document were presented at the recent NASA workshop in Rohnert Park July 13-15th, 2007. We have been preparing to go to the NSRL for iron ions (September 2007, NSRL-run7C). The cells we have employed for these tasks are human NTerra2/D1 can be induced to differentiate into mature neurons and astrocytes were given to us by Marcelo Vasquez. The original five week induction protocol yielded neuronal and astrocytic cells that had a mixed phenotype with respect to the markers they expressed suggesting that the induction protocol was not producing mature or pure populations of the two cell types of interest. To improve the outcome a newer protocol was used to induce the differentiation of the precursor stem cells. The most recent set of cells appear to be purer and better separated into their respective populations. The improved 5 week protocol has been used to generate the cultures used for all experiments discussed here in. Since the initiation of our grant (October 1st, 2006) we have completed our gamma irradiation series using our neurons, astrocytes and physically mixed cultures. These studies not only provide a point of reference they also allow us to work out the details and logistics prior to conducting these experiments at BNL. The endpoints were; function (3H-glutamate uptake); apoptosis (annexin V binding and BrdU incorporation); glutamate transporter expression (immunocytochemical quantification and Western blot analysis). These endpoints constitute the same endpoints required for the completion of

Task Progress:	aim 1 with HZE particles. We have also worked out the assays for measuring reactive oxygen species levels and antioxidants which will be required for aim 2; as well as, perfected microinjection and purinergic receptor expression (required for aim 3). We have also conducted a proton experimental series to complement our gamma-ray data. We have plans to obtain a biological replicate proton experiment in mid-to late October, 2007. We have also continued to generate isolated cultures of neurons and astrocytes to archive for achieving all of the planned experiments. Results to date for aim 1 using Carbon ion irradiation: Upon return from the NSRL run in March 2006 we have worked to complete the carbon irradiation endpoints constituting aim 1. Currently we have processed the functional assay results, apoptosis assays and quantification of the glutamate transporters 1-4, as well as, the purinergic receptors (part of aim 3). As we found with gamma- irradiation the glutamate uptake was reduced in astrocytes exposed to 50 & 200cGy at 3 hours, returning to basal by 2 days. Gamma-irradiated neurons were induced to take-up glutamate, with an increase measured at 3hours persisting throughout the 7day time point. Apoptosis (Annexin V binding) following carbon radiation revealed neurons were the most sensitive to this measure of damage. The signal was linear and increased over time. Whereas, in the astrocytes and mixed cultures the levels of binding were relatively low. In contrast to the Annexin V data, the BrdU-incorporation yielded results for 3 hours that were similar for astrocytes and neurons, however at 2 days the astrocytes had significantly more 3'-end breaks than the neurons. In the mixed cultures there was a muted response with little increase in BrdU-incorporation over time or dose. The functional assessment post carbon ion exposure was similar to the results found following gamma-irradiation; although the changes were seen at all doses and persisted through the 2day time point. The levels of the glutamate transporters fluctuated over the time and dose range similar to that following gamma-irradiation. The significance of the moderate changes in the transporter levels awaits further investigation. Conclusion to date: The response of NTerra-2/D1 derived astrocytes and neurons to carbon ions is similar to the results obtained from gamma-irradiation except that the responses were more severe in both magnitude and duration. We had not previously conducted extensive physically mixed experiments with gamma-rays so there is was no direct comparison. However, the mixed culture responses for carbon were surprisingly muted compared to either cell type alone. This suggests that physical coupling between the two cell types provides some measure of protection. The data shown for carbon ions are from one experiment (March 2007-NSRL7A) a repeat set of experiments is scheduled for the spring of 2008. This task is systematically undertaking the aims as listed below. The goal of this project is to determine whether exposure to doses and radiation qualities relevant to space will result in detrimental effects on neurological components of the CNS. With that in mind the immediate plans are to complete the various phases of the project and publish the results. This project of Dr. Green's involves one of her graduate students, Martha Sanchez whose is currently receiving a student stipend from our NASA grant. Martha Sanchez attended the second annual NASA-sponsored summer school and is an excellent student who will undoubtedly contribute to space-relevant research now and in her future career. Martha is expected to receive her PhD in the Fall of 2007.
Bibliography Type:	Description: (Last Updated: 01/29/2015)
Abstracts for Journals and Proceedings	Sanchez MC, Benitez A, Ortloff LS, Green LM. "Glutamate transporter response after exposure to low-dose gamma and proton radiation." International Congress Radiation Research, San Francisco, CA, July 8th-12th, 2007. International Congress Radiation Research, San Francisco, CA, July 8th-12th, 2007. Received SIT award. , Jul-2007
Abstracts for Journals and Proceedings	Sanchez MC, Bianski BM, Green LM. "Response of Astrocytes and Neurons to Carbon Ions." 18th Annual NASA Space Radiation Investigators' Workshop, Rohnert Park, CA, July 13-15, 2007. 18th Annual NASA Space Radiation Investigators' Workshop, Rohnert Park, CA, July 13-15, 2007. Also received NASA travel award to cover costs of student participation. , Jul-2007